

Programme	BSc Statistics				
Course Code	STA3MN202 (P)				
Course Title	Statistical inference for data science				
Type of Course	Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Thorough knowledge in probability concept and Random variables.				
Course Summary	Students will possess a wide understanding of Law of Large Numbers, ANOVA, and non-parametric tests and they will be equipped to apply these statistical techniques to various scenarios, making informed decisions and drawing meaningful conclusions from data.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Introduce and explore different law of large numbers	U	C	Instructor-created exams / Quiz
CO2	Define and understand the rationale for testing differences between two populations	R	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Formulate null and alternate hypothesis for a real life two population problem and analyze data to help entrepreneurial decisions using critical thinking skills.	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Differentiate between one-way and two-way ANOVA and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Instructor-created exams / Home Assignments
CO5	Define and understand the principles of non parametric statistics	U	F	One Minute Reflection Writing assignments/

				Instructor-created exams
CO6	Describe analysis of variance and hypothesis testing using R software.	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Units	Content	Hrs (45 +30)	Marks (70)
I	LAW OF LARGE NUMBERS		10	15
	1	Chebychev's inequality (Definition and Problems)	2	
	2	Modes of Convergence of a Sequence of Random Variables	1	
	3	Weak Law of Large Numbers (Statement and Problems)	2	
	4	Bernoulli's Law of Large Numbers		
	5	Strong Law of Large Numbers	2	
	6	CLT (Lindeberg- Levy)	3	
	Sections from References: Unit 1: 7.5 [Ref 1] Unit 2: 7.6 [Ref 1] Unit 3: 7.7 [Ref 1] Unit 4:7.7 [Ref 1] Unit 5: 7.8[Ref 1] Unit 6: 9.16[Ref 1]			
II	HYPOTHESIS TESTING: TWO POPULATIONS		12	20
	6	Test of Significance for difference of two population proportions (Concept and Problems)	2	
	7	Test of Significance for difference of two population means (Large Sample-Concept and Problems)	2	
	8	Test of Significance for difference of two population means (Small Sample-Concept and Problems)	3	

	9	Paired t test(Concept and Problems)	3	
	10	F test for equality of proportions	2	
	Sections from References: Unit 6: 14.7.2 [Ref 1] Unit 7: 14.8.4 [Ref 1] Unit 8: 16.3.2 [Ref 1] Unit 9: 16.3.3[Ref 1] Unit 10: 16.7[Ref 1]			
III	ANALYSIS OF VARIANCE		8	15
	11	ANOVA	1	
	12	One-Way Analysis of Variance	3	
	13	Two -Way Analysis of Variance	4	
	Sections from References: Unit 11: 23.1,23.2 [Ref 2] Unit 12: 23.3[Ref 2] Unit 13: 23.4[Ref 2]			
IV	NON PARAMETRIC TEST		15	20
	14	Introduction to Non Parametric Methods	1	
	15	Advantages and Limitations	1	
	16	Sign Test- one sample	3	
	17	Wilcoxon Signed Rank Test	2	
	18	Mann- Whitney Test	2	
	19	Kruskal- Wallis Test	2	
	20	Single Sample Run Test	2	
	21	Median Test	2	
	Sections from References: Unit 14: 26.2[Ref 2] Unit 15:26.2.1 [Ref 2] Unit 16: 26.3[Ref 2] Unit 17: 26.4 [Ref 2] Unit 18:26.5 [Ref 2] Unit 19:26.7[Ref 2] Unit 20: 26.8[Ref 2] Unit 21: 26.9[Ref 2]			

V	PRACTICUM		30	
	Do practice problems in R software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Plots to check normality		
	2	Hypothesis testing		
	3	Goodness of fit tests		
	4	Correlation		
	5	Inference procedures for correlation coefficient		
	6	Linear regression		
	7	Inference procedures for simple linear model		
	8	Polynomial regression models		
	Sections from References: Unit 1: 4.4 [Ref 5] Unit 2: 4.5 [Ref 5] Unit 3: 4.6 [Ref 5] Unit 4: 5.2 [Ref 5] Unit 5: 5.3 [Ref 5] Unit 6: 5.4 [Ref 5] Unit 7: 5.5 [Ref 5] Unit 8: 5.8 [Ref 5]			
Books and References: 1. Gupta, S. C. and Kapoor, V. K. (2002). Fundamentals of Mathematical Statistics. , 11 th edition, Sulthan Chand, New Delhi. 2. Gupta, S. C. (2015). Fundamentals of Statistics,7 th Edition ,Himalaya Publishing House. 3. Gupta S.C (2021), Statistical Methods, 46 th edition, Sultan Chand and Sons. 4. Prem S. Mann (2016), Introductory Statistics 9 th Edition ,Wiley 5. W. N. Venables, D. M. Smith and the R Core Team, An Introduction to R (2023) 6. Sudha G Purohith, Sharad D Core, Shailaja R Deshmukh (2015), Statistics Using R				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	2	-	-	-	3	1	2	-	-	-	-
CO 2	2	-	-	-	-	2	2	1	-	-	-	-
CO 3	-	-	2	-	-	2	2	-	-	1	3	-
CO 4	2	-	-	-	-	-	2	1	-	-	-	3
CO 5	-	3	-	2	3	3	-	2	1	-	-	-
CO 6	-	-	-	-	-	2	3	1	-	-	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5	✓	✓		✓
CO 6	✓			